FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

Department of Computer Engineering
Academic Year 2025-26

Rubrics for Lab Experiments

Class : *B.E. Computer Engineering*

Subject Name :*BDA Lab*

Semester : VII

Subject Code :CSL702

Practical No:	5
Title:	**Write a command to perform insert, create, update and delete Cassandra**
Date of Performance:	05-08-2025
Roll No:	9913
Name of the Student:	Mark Lopes

Evaluation:

Performance Indicator	Below average	Average	Good	Excellent	Marks
On time Submission (2)	Not submitted (0)	Submitted after deadline (1)	Early or on time submission(2)	---	
Test cases and output (4)	Incorrect output (1)	The expected output is verified only a for few test cases (2)	The expected output is Verified for all test cases but is not presentable (3)	Expected output is obtained for all test cases. Presentable and easy to follow (4)	
Coding efficiency (2)	The code is not structured at all (0)	The code is structured but not efficient (1)	The code is Structured and efficient. (2)	-	
Knowledge(2)	Basic concepts not clear (0)	Understood the basic concepts (1)	Could explain the concept with suitable example (1.5)	Could relate the theory with real world application(2)	
Total					

Signature of the Teacher:

Experiment No 5

Aim: Write a program to insert, search, update, delete and aggregate data using MongoDB
NoSQL Database

Objective:

- Understand NoSQL databases architecture
- Understand commands to insert document using MongoDB
- Understand commands to search document using MongoDB
- Understand commands to update document using MongoDB
- Understand commands to delete document using MongoDB
- Understand commands to implement aggregates using MongoDB

Input& Output:

Commands to install mongodb
$Ps –ef|grep mongo
$Sudo apt-get update
$Sudo apt-get install mongodb
$Sudo service mongod start
$mongoimport –db test –collection restaurants --drop --file /home/fcrit/Desktop/primer-
dataset.json
Commands to create documents using mongodb
$mongo
>show db
>show collections
> use test
>db.restaurants.insert()
db.restaurants.insert(
{
"address": {
"street":"2 Avenue",

"zipcode":"10075",
"building":"1480",
"coord": [-73.9557413, 40.7720266],
},
"borough":"Manhattan",
"cuisine":"Italian",

```
&quot;grades&quot;: [
{
&quot;date&quot;:ISODate(&quot;2014-10-01T00:00:00Z&quot;),
&quot;grade&quot;:&quot;A&quot;,
&quot;score&quot;:11
},
{
&quot;date&quot;:ISODate(&quot;2014-01-16T00:00:00Z&quot;),
&quot;grade&quot;:&quot;B&quot;,
&quot;score&quot;:17
}
],
&quot;name&quot;:&quot;Vella&quot;,
&quot;restaurant_id&quot;:&quot;41704620&quot;
}
)
&gt;db.restaurants.find()
{&lt;field1&gt;:&lt;value1&gt;,&lt;field2&gt;:&lt;value2&gt;,...}
db.restaurants.find({&quot;borough&quot;:&quot;Manhattan&quot;})
db.restaurants.find({&quot;grades.score&quot;:{$gt:30}})
db.restaurants.find({&quot;grades.score&quot;:{$lt:10}})
```

Logical AND

```
db.restaurants.find({&quot;cuisine&quot;:&quot;Italian&quot;,&quot;address
.zipcode&quot;:&quot;10
075&quot;})
```

Logical OR

```
db.restaurants.find(
{$or:[{&quot;cuisine&quot;:&quot;Italian&quot;},{&quot;address.zipcode&qu
ot;:&quot;10075&quot;}]}
)
```

Sort query result

```
db.restaurants.find().sort({&quot;borough&quot;:1,&quot;address.zipcode&q
uot;:1})
&gt;db.restaurants.update()
db.restaurants.update(
{&quot;name&quot;:&quot;Juni&quot;},
{
$set:{&quot;cuisine&quot;:&quot;American (New)&quot;}
}
)
```

Multiple documents

db.restaurants.update(
{"address.zipcode":"10016",cuisine:"Other&quo
t;},
{
$set:{cuisine:"Category To Be Determined"},
$currentDate:{"lastModified":true}
},
{multi:true}
)
>db.restaurants.remove()
db.restaurants.remove({"borough":"Manhattan"})
db.restaurants.remove({"borough":"Queens"},{justOn
e:true})
db.restaurants.remove({})
Drop table
db.restaurants.drop()
Result: Implementation of the solution for above problem and attach
commands with output

Experiment No 5
Aim: Write a command to perform insert, create, update and delete Cassandra
(NoSQL)
database

● Understand NoSQL databases architecture
● Understand commands to create database using cassendra
● Understand commands to insaert data using cassendra
● Understand commands to update data using cassendra
● Understand commands to delete data using cassendra

Steps for creating database:
Using Docker run Cassandra commands:
Commands for simple database creation:

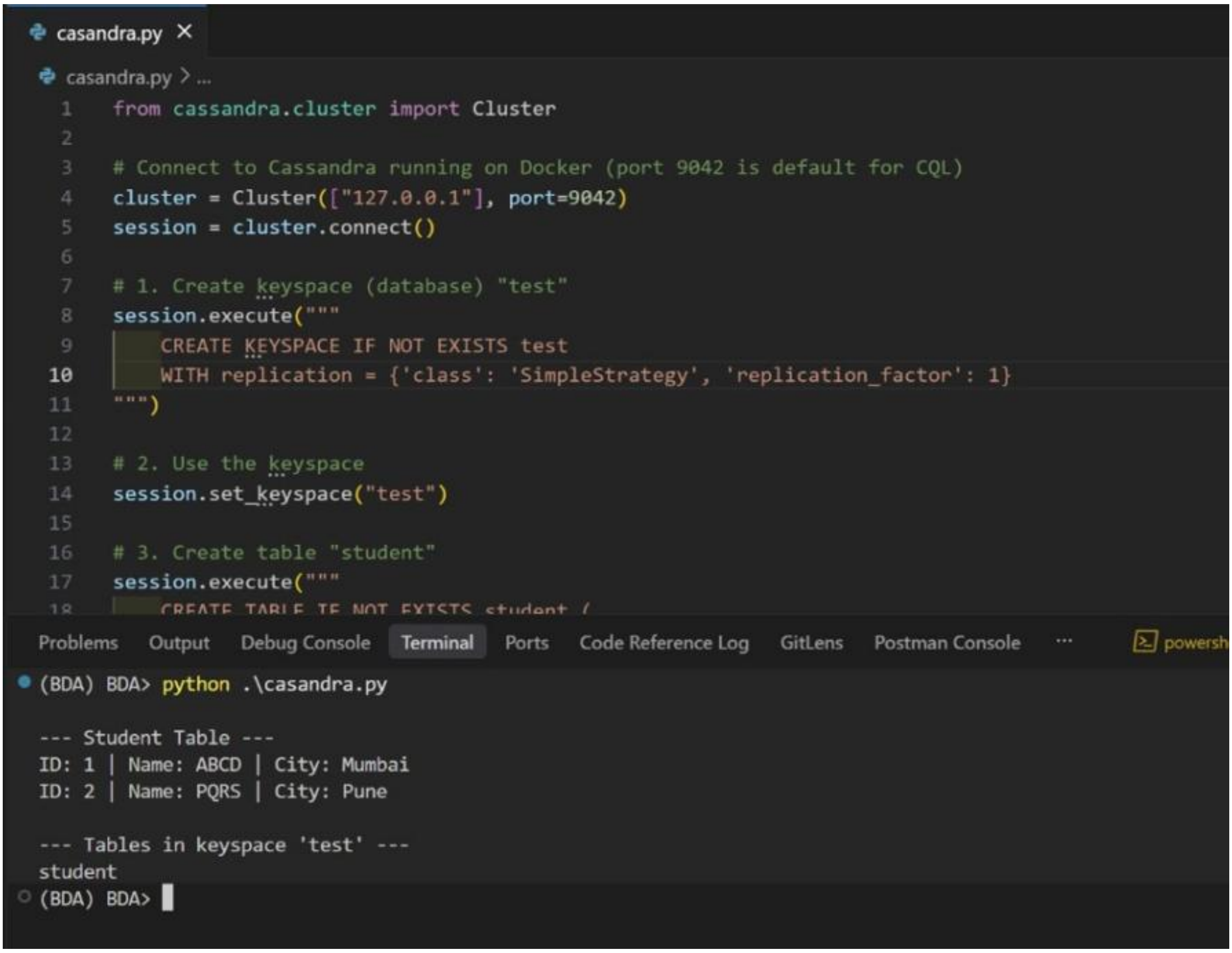
1. Sudo apt install docker.io
2. sudo docker pull Cassandra:latest
3. sudo docker run –d – name cassandra-node –p 9042:9042 cassandra
(note: 9042 port reserverd for Cassandra)
4. sudo docker exec –it cassandra-node bash
5. cqlsh (note: cqlsh->Cassandra query language shell)
6. create keyspace test with replication={‘class’: ‘SimpleStrategy’,

‘replication_factor’:1};
(note:keyspace means database,here name of database is test)

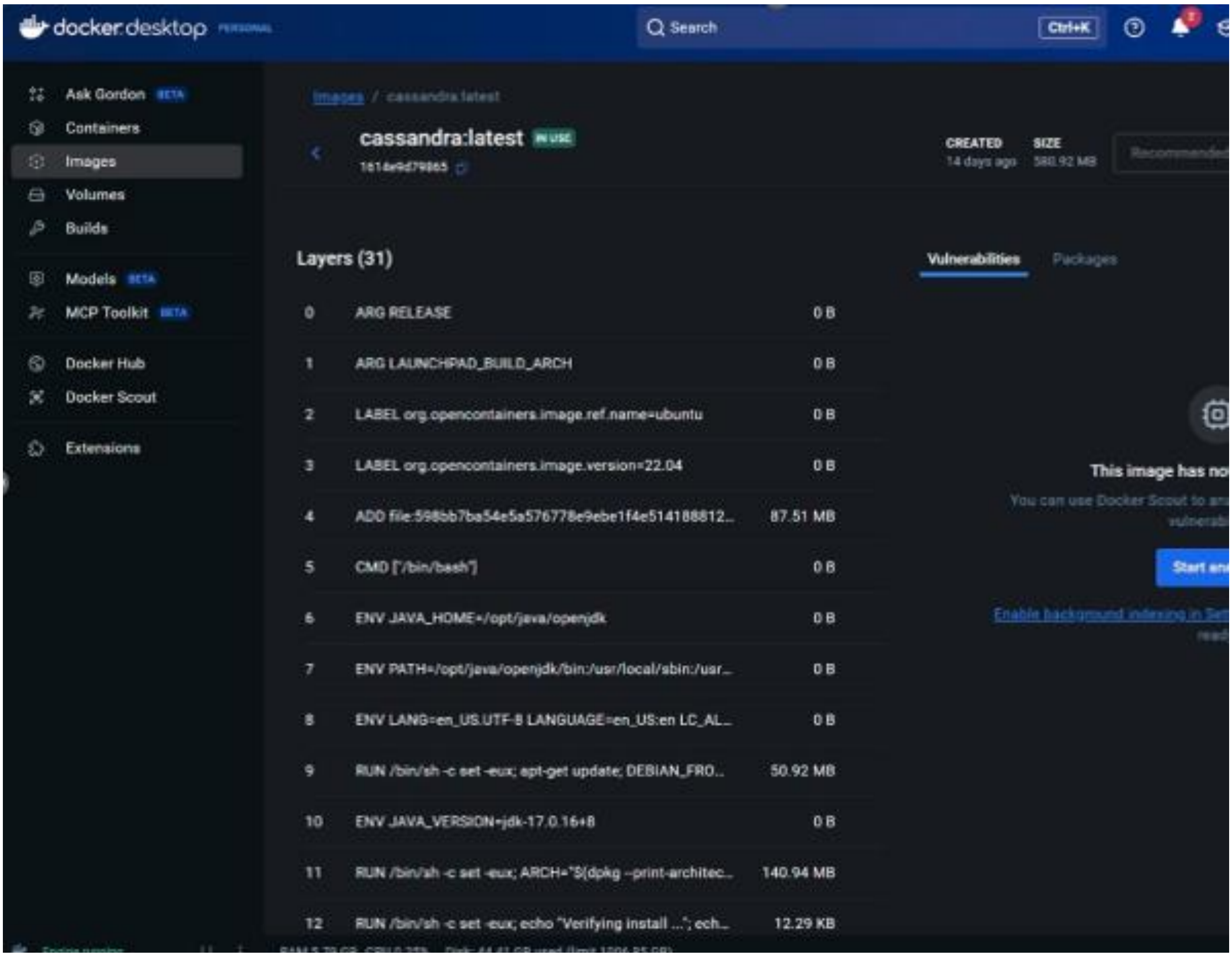
7. use test;
8. create table student(s_id int primary key,s_name text,s_city text);
9. insert into student(s_id,s_name,s_city)values(1,’ABCD’,’Mumbai’);
9. insert into student(s_id,s_name,s_city)values(2,’PQRS’,’Pune’);
10. select * from student;

Result: Implementation of the solution for above problem and attach commands with output

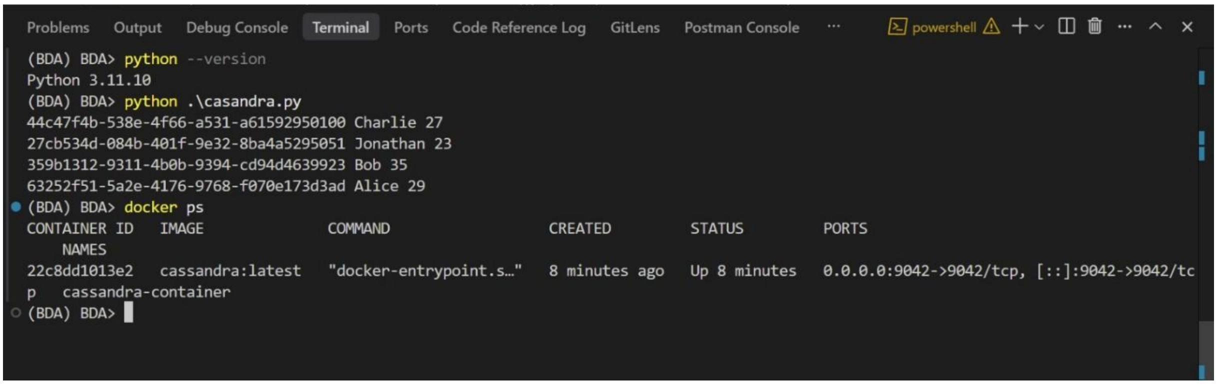
Conclusion:



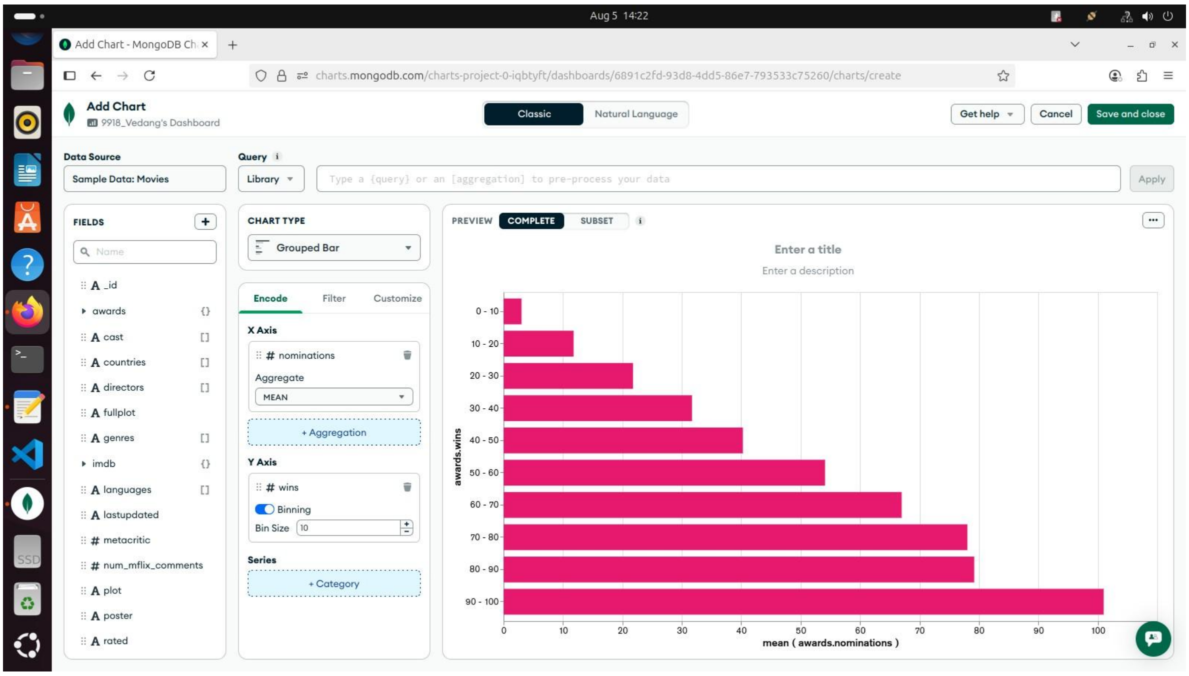
```
(BDA) BDA> docker exec -it cassandra-container  bash
root@22c8dd1013e2:/# cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.5 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE IF NOT EXISTS test
n = {'class': 'SimpleStrategy', 'replication_facto   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE test;
cqlsh:test> CREATE TABLE IF NOT EXISTS student (
        ...     s_id int PRIMARY KEY,
        ...     s_name text,
        ...     s_city text
        ... );
cqlsh:test> INSERT INTO student (s_id, s_name, s_city) VALUES (1, 'shadv', 'Hyderabad');
cqlsh:test> INSERT INTO student (s_id, s_name, s_city) VALUES (2, 'gosdb', 'Goa');
cqlsh:test> SELECT * FROM student;

 s_id | s_city    | s_name
------+-----------+--------
    1 | Hyderabad |  shadv
    2 |       Goa |  gosdb
```




```
cqlsh:test> DESCRIBE TABLES;

student

cqlsh:test> DESCRIBE KEYSPACES;

system       system_distributed  system_traces  system_virtual_schema  testks
system_auth  system_schema       system_views   test

cqlsh:test>
```


Problems Output Debug Console Terminal Ports Code Reference Log GitLens Postman Console
powershell
(BDA) BDA> python --version
Python 3.11.10
(BDA) BDA> python .\casandra.py
44c47f4b-538e-4f66-a531-a61592950100 Charlie 27
27cb534d-084b-401f-9e32-8ba4a5295051 Jonathan 23
359b1312-9311-4b0b-9394-cd94d4639923 Bob 35
63252f51-5a2e-4176-9768-f070e173d3ad Alice 29
(BDA) BDA> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
22c8dd1013e2 cassandra:latest "docker-entrypoint.s…" 8 minutes ago Up 8 minutes 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tc
p cassandra-container
(BDA) BDA>

PART-B


Aug 5 14:22
Add Chart - MongoDB Ch
charts.mongodb.com/charts-project-0-iqbtyft/dashboards/6891c2fd-93d8-4dd5-86e7-793533c75260/charts/create
Add Chart
9918_Vedang's Dashboard
Classic
Natural Language
Get help
Cancel
Save and close
Data Source
Sample Data: Movies
Query
Library
Type a {query} or an [aggregation] to pre-process your data
Apply
FIELDS
_id
awards
cast
countries
directors
fullplot
genres
imdb
languages
lastupdated
metacritic
num_mflix_comments
plot
poster
rated
CHART TYPE
Grouped Bar
Encode
Filter
Customize
X Axis
nominations
Aggregate
MEAN
+ Aggregation
Y Axis
wins
Binning
Bin Size 10
Series
+ Category
PREVIEW
COMPLETE
SUBSET
Enter a title
Enter a description
0 - 10
10 - 20
20 - 30
30 - 40
40 - 50
50 - 60
60 - 70
70 - 80
80 - 90
90 - 100
awards.wins
0 10 20 30 40 50 60 70 80 90 100
mean (awards.nominations)